Food Analytical Methods (2026) 19:158
https://doi.org/10.1007/s12161-026-03067-y

RESEARCH

An Eco-friendly Magnetic Solid-Phase Extraction Method for the Determination of Pendimethalin and Fenarimol Pesticides in Fruit Juices by HPLC-DAD

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Received: 17 November 2025 / Accepted: 3 March 2026 / Published online: 13 March 2026
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Abstract
Ensuring food safety necessitates the development of sensitive, reliable, and environmentally sustainable analytical methods for pesticide residue determination. An efficient chromatographic approach was developed for trace quantification of pendimethalin and fenarimol by combining a newly synthesized magnetic solid-phase extraction (MSPE) sorbent, Fe_3O_4@APTES-uric acid, with high-performance liquid chromatography coupled to diode array detection (HPLC-DAD). Critical extraction parameters, including sample pH, elution solvent type and volume, and desorption conditions, were systematically optimized to maximize recovery and analytical performance. Under optimized conditions, fenarimol and pendimethalin were monitored at 202 nm and 239 nm, respectively. The method showed excellent sensitivity, with limits of detection of 3.22 ng mL^{-1} for fenarimol and 3.32 ng mL^{-1} for pendimethalin. The applicability of the proposed MSPE–HPLC-DAD method was successfully demonstrated through the analysis of commercial fruit juice samples, yielding satisfactory recovery values and confirming the method's accuracy and robustness. Good precision was obtained via relative standard deviation values below 5.8% for target molecules. The practical applicability and environmental impact of the method were evaluated using the Blue Applicability Grade Index (BAGI) and the Metric for Green Analytical Procedures Index (MoGAPI). The method achieved scores of 72.5 and 80, respectively, reflecting good suitability for routine analysis and high environmental performance.

Keywords Magnetic solid-phase extraction (MSPE) · High-performance liquid chromatography (HPLC-DAD) · Pesticide residues · Pendimethalin · Fenarimol · Food safety

Introduction

Pests are broadly recognized as organisms that disrupt agricultural productivity, ecological stability, or human well-being (Sidhu et al. 2019). This group includes insects, fungi, bacteria, and other biological agents capable of reducing crop yields, damaging livestock, and ultimately threatening food security (Oerke 2006). Their uncontrolled presence not only causes significant economic losses but also undermines the ecological balance of natural systems (Pimentel 2005). To mitigate these impacts, pesticides have been developed as chemical formulations specifically designed to suppress, control, or eliminate harmful species. They are extensively applied in agriculture, urban environments, and public health initiatives to manage a wide range of pests, including weeds, insects, fungi, and bacterial pathogens (Cooper and Dobson 2007). By selectively targeting these organisms, pesticides play a critical role in crop protection, productivity enhancement, and sustaining global food supply chains (Pretty and Bharucha 2015). Among widely used pesticides, pendimethalin and fenarimol stand out as two major compounds due to

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their proven effectiveness in controlling weeds and fungal pathogens (Bartlett et al. 2002).

Pendimethalin, chemically identified as N-(1-ethylpropyl)−3,4-dimethyl-2,6-dinitrobenzenamine, is a substituted aniline derivative characterized by nitro substituents at the 2 and 6 positions and methyl groups at the 3 and 4 positions of the aromatic ring (Acar et al. 2022; Wang et al. 2023). Functioning primarily as a pre-emergent herbicide, it inhibits weed seed germination before emergence, thereby ensuring effective early-stage crop protection (Łozowicka et al. 2013). It is widely applied in cereals, legumes, maize, soybeans, and various vegetable crops and is also valued in ornamental plant cultivation (Strandberg and Scott-Fordsmand 2004). Fenarimol, chemically defined as [2-chlorophenyl] [4-chlorophenyl] pyrimidin-5-ylmethanol, is a systemic fungicide belonging to the pyrimidine class. Its primary mode of action involves disrupting ergosterol biosynthesis, a key component of fungal cell membranes, which effectively prevents spore germination and fungal propagation (European Food Safety Authority et al. 2023). Fenarimol has been extensively used in the management of fruit and vegetable crops, particularly against powdery mildew and other economically important fungal diseases (Karsauliya et al. 2021). Together, pendimethalin and fenarimol represent indispensable components of integrated pest management strategies. Their combined use not only enhances crop productivity but also contributes to agricultural sustainability and food safety (Pimentel 2005). Chemical formulas and properties of pendimethalin and fenarimol are present in Table 1.

Sample preparation is a critical step in analytical chemistry, involving the separation of trace analytes from complex matrices and their concentration into a smaller volume. This process enhances the analyte's concentration, facilitating accurate detection and quantification (Nenni et al. 2022; HI. Ulusoy et al. 2023). Samples such as food, environmental samples (e.g., water and soil), biological fluids (e.g., blood, serum, and urine), and pharmaceuticals contain not only the analyte but also various interfering substances. The complete set of chemical and physical components in a sample, including the target analytes, co-extracted compounds, and particulates, constitutes the sample matrix. This matrix can significantly interfere with the isolation and accurate quantification of analytes, making effective separation a prerequisite for reliable analysis (Poole 2003; Risticevic et al. 2010).

Conventional sample preparation techniques, such as liquid-liquid extraction (LLE) and solid-phase extraction (SPE), are designed to address this challenge by removing interferences and pre-concentrating the analyte, thereby improving the sensitivity and accuracy of the analysis. Among these, SPE has become a cornerstone technique due to its higher efficiency, reduced solvent consumption, and greater user-friendliness compared to LLE (Buszewski and Szultka 2012; Demir et al. 2023; Karaca et al. 2020). Building upon the principles of SPE, magnetic solid-phase extraction (MSPE) has emerged as a powerful advancement. MSPE utilizes functionalized magnetic nanoparticles as sorbents, which can be easily dispersed into a sample and rapidly retrieved using an external magnetic field. This design eliminates the need for time-consuming centrifugation or filtration steps, simplifying the procedure and minimizing potential analyte loss (Capriotti et al. 2019; Vasconcelos and Fernandes 2017). The significance of MSPE lies in its ability to provide a rapid, efficient, and environmentally friendly platform for extracting trace analytes from complex biological, food, and environmental samples. Its key advantages include a high surface area for extraction, tunable surface chemistry for selectivity, excellent sorbent reusability, and a substantial reduction in the volume of organic solvents required. This makes MSPE highly compatible with modern chromatographic and spectrometric techniques and versatile for targeting diverse analytes, including pesticides, pharmaceuticals, and heavy metals, through specific nanomaterial functionalization (Hou et al. 2025; Zhang et al. 2018).

The primary objective of this study is to develop, optimize, and validate a novel analytical method for the simultaneous determination of pendimethalin and fenarimol in complex matrices such as fruit juice. The fundamental innovation is based on the synthesis and application of Fe_3O_4@

Table 1 The physical and chemical properties of pendimethalin and fenarimol molecules

Parameter	Pendimethalin	Fenarimol
Pesticide Type	Herbicide	Fungicide
Molecular structure		
Chemical Formula	$C_{13}H_{19}N_3O_4$	$C_{17}H_{12}Cl_2N_2O$
IUPAC name	N-(1-ethylpropyl)−3,4-dimethyl-2,6-dinitroaniline	2-(4-chlorophenyl)−4-(chlorophenyl)-pyrimidin-5-ylmethanol
Molecular weight (g/mol)	281.31	331.2
Solubility in water (mg/L, 20 C)	0.33	13.70
logP(Octanol–Water)	5.40	3.69
pKa (25 ^{0}C)	2.80	2.58

APTES-uric acid, a novel magnetic solid-phase extraction (MSPE) sorbent not previously reported in the literature. Uric acid was strategically chosen as a functional ligand to provide numerous hydrogen donor/acceptor and π-π interaction sites on the magnetic support surface. This enabled the selective separation and efficient pre-concentration of the target pesticides. The developed MSPE method, combined with an HPLC-DAD system, offers a selective, highly recoverable, accurate, sensitive, matrix effect—minimizing, fast, reliable, and environmentally friendly analytical approach for the determination of these contaminants in complex food samples. The environmental sustainability and practical applicability of the proposed method were investigated using current green chemistry and applicability metrics. Moreover, the developed method was assessed not only analytically but also in terms of environmental and practical performance using MoGAPI and BAGI indexing tools.

Materials and Methods

Instrumentation

All solutions were prepared using ultrapure water with a resistivity of 18.2 MΩ·cm, generated by an ELGA Pure Lab Flex III system. Chromatographic measurements were conducted using a Shimadzu Prominence HPLC system (Kyoto, Japan). The system comprised a quaternary pump (LC-20AD), a photodiode array detector (SPD-M20A PDA), a vacuum degasser (DGU-20A), a column oven (CTO-10AS VP), and an autosampler (SIL-20AC). Separation and analysis were performed on a reverse-phase C18 column (Inertsil C18, 250 mm × 4.6 mm, 5 µm), and chromatograms were processed using LC Solution 2.0 software (Shimadzu). Prior to analysis, all solvents were filtered through a 0.45-µm PTFE membrane filter (Millipore, HNWP) and degassed in an ultrasonic bath for 10 min (JP Selecta, Spain).

Fourier transform infrared (FTIR) spectra were recorded with a Perkin-Elmer Spectrum 400 spectrometer (Waltham, MA, USA). X-ray diffraction (XRD) analysis was performed using a Bruker AXS D8 diffractometer, while Raman spectroscopy was carried out with a Renishaw 1000 microspectrometer employing a 514.5 nm excitation wavelength. The morphology and structural characteristics of the materials were examined using scanning electron microscopy (LEO 440 SEM). These instruments and procedures ensured precise, reproducible, and reliable characterization and analysis of the synthesized materials and chromatographic samples.

Chemicals and Reagents

All chemicals used in this study were of analytical grade (≥ 99% purity) and obtained from Sigma-Aldrich or Merck. Pendimethalin and fenarimol pesticides were bought from Sigma-Aldrich (St. Louis, MO). HPLC solvents including methanol and acetonitrile were obtained from Merck company. Iron salts ($FeCl_3 \cdot 6H_2O$, $FeCl_2 \cdot 4H_2O$), aminopropyl triethoxysilane (APTES), and uric acid were obtained from Sigma-Aldrich (St. Louis, MO). Stock solutions were prepared by dissolving the pure pesticides in methanol. Calibration standards for HPLC analysis were prepared by diluting the stock solutions. All solutions were stored in amber glass bottles at 4 °C to prevent light-induced degradation.

A Britton-Robinson (BR) Buffer system was used in the range of pH 2.0–10.0, including 0.04 M of boric acid, phosphoric acid, and acetic acids. The mixture was stirred thoroughly and diluted to 1 L in a volumetric flask. Buffer solutions at specific pH values were adjusted according to the respective pKa values of the constituent acids and verified with a calibrated pH meter. Prepared buffers were stored in conditions protected from light and heat to maintain stability.

Synthesis of Solid Phase Support Material, Fe_3O_4@ APTES-Uric Acid

A well-established synthesis method, previously employed in our studies and extensively reported in the literature (Hou et al. 2025; S. Ulusoy et al. 2022, 2024; Wierucka and Biziuk 2014), was utilized for the preparation of functionalized magnetic nanoparticles. Initially, 10.8 g of $FeCl_3 \cdot 6H_2O$ and 3.98 g of $FeCl_2 \cdot 4H_2O$ were dissolved in a mixture containing 50 mL of 0.1 M HCl and 60 mL of ethanol:water (1:1) solution. The solution was stirred at 55 °C and 600 rpm under a nitrogen atmosphere to maintain inert conditions. Subsequently, 30 mL of ammonia was added dropwise to induce the formation of magnetite (Fe_3O_4) nanoparticles. The resulting black suspension was subjected to an external magnetic field to isolate the Fe_3O_4 particles, which were then washed multiple times with a water–ethanol mixture and dried at 60 °C for 6 h.

The surface modification was performed in two consecutive stages to obtain the final functional sorbent. First, APTES silanization was conducted to introduce primary amine groups onto the magnetite surface. For this purpose, 3.0 g of the dried Fe_3O_4 particles was dispersed in 100 mL of ethanol: water (1:1) solution. To catalyze the silanization reaction, 3.0 mL of 6 M NH_3 ammonia was added to the mixture on a magnetic stirrer at 500 rpm and then followed by 3 mL of aminopropyl triethoxysilane (APTES) to coat the particle surfaces. After 12 h, 300 mg of uric acid was dissolved in 50 mL of methanol:water(1:1) and added to the mixture for surface modification. The final product, Fe_3O_4@ APTES-uric acid, was isolated magnetically, washed extensively with the ethanol-water mixture to remove any unreacted reagents, and air-dried at room temperature for 24 h.

HPLC Conditions for Determination of Fenarimol and Pendimethalin Molecules

Before conducting the MSPE experiments, the HPLC conditions for the direct determination of fenarimol and pendimethalin pesticides using HPLC were optimized step by step. The used HPLC method has been developed and partly validated for the first time in this study. Preliminary trials were performed with various columns as the stationary phase, and the Inertsil C18 column was identified as the most suitable. To determine the ideal mobile phase composition, numerous trials were conducted using different aqueous solutions containing organic mobile phases such as methanol, ethanol, and acetonitrile, combined with buffers at various pH levels. These trials were carried out in both isocratic and gradient elution modes. The experiments continued until the most distinct peaks for fenarimol and pendimethalin were achieved.

The ideal HPLC working conditions obtained because of the optimization process are provided in Table 2.

The Proposed MSPE-HPLC-DAD Method

Fifty milligrams of magnetic sorbent (Fe_3O_4@APTES-uric acid) was weighed and transferred to 50-mL falcon tubes. Ten milliliters of distilled water and 2 mL of pH 4.0 buffer were added to the tubes. Then, 10 mL of the sample solution containing both pesticide types (fenarimol and pendimethalin) in the range studied (10.0–1500.0 ng mL^{-1}) was added, and the volume of the tube was completed to 50 mL with distilled water. The caps of the falcon tubes were tightly closed and placed on the rotator device. The device operated at 50 rpm for 40 min. After the time was up, the tubes were placed on the magnetic apparatus, and the aqueous phase was separated with the help of a pipette. After separating the aqueous phase, 500 µL of methanol:acetonitrile (1:1) (back extraction or desorption) was added to the solid phase, and the tubes were vortexed for 60 s. The aim of this process was to separate the analyte from the solid phase and transition to the solvent phase. Solvent liquid phase samples were taken into the injector with the help of a magnetic field, passed through a 0.45-µm injector tip filter, transferred to vials and placed in the HPLC-DAD device. In this way, the enriched fenarimol and pendimethalin samples are determined by HPLC-DAD device.

Table 2 Optimized HPLC-DAD conditions for the determination of fenarimol and pendimethalin

Parameter	Finding
HPLC mode	Isocratic elution
Mobil phase (eluent)	30% Methyl alcohol 40% Trifluoro acetic acid (0.1%, (v/v)) 30% Acetonitrile
Eluent flow rate	1 mL/min
Run-time	15 min
Column	Inertsil C 18, (250 mm × 4.6, 5.0 µm)
Column temperature	35 °C
Injection volume	10 µL

Results and Discussions

Characterization Used Magnetic Sorbent [Fe_3O_4@APTES-Uric Acid]

SEM Analysis

The SEM micrographs clearly illustrate the morphological differences between core Fe_3O_4 nanoparticles (Fig. 1A) and the surface-modified Fe_3O_4@APTES-uric acid material (Fig. 1B). In the case of Fe_3O_4, the surface appears relatively compact with irregularly shaped aggregates and small spherical nanoparticles distributed across the matrix (Fig. 1A). The particles exhibit a somewhat heterogeneous size distribution, with evidence of clustering and partial agglomeration, which is typical for bare magnetic nanoparticles due to their strong magnetic dipole–dipole interactions. In contrast, after the APTES-assisted silica coating and subsequent uric acid functionalization, the surface morphology undergoes a remarkable transformation. The Fe_3O_4@APTES-uric acid sample (Fig. 1B) shows the presence of well-defined spherical structures with smoother surfaces compared to the unmodified Fe_3O_4. The coating process appears to reduce irregularities and produce a more uniform morphology, while still maintaining a porous and rough background that can enhance surface area. This structural modification is a strong indication of the successful deposition of silica layers, providing a stable framework for subsequent grafting of organic ligands such as uric acid.

The morphological evolution from irregular Fe_3O_4 aggregates to more uniform and functionalized microspheres demonstrates the efficiency of the modification process. The presence of spherical domains in Fig. 1B suggests improved colloidal stability, which is advantageous for adsorption-based applications. Furthermore, the smoother surfaces combined with the chemical affinity of uric acid are expected to provide additional active sites for pesticide capture, thereby enhancing the overall performance of the adsorbent. In summary, the SEM images confirm the morphological progression associated with the functionalization strategy, highlighting the successful transition from raw Fe_3O_4 to a more structured and chemically versatile Fe_3O_4@APTES-uric acid system. This supports the material's potential in pesticide adsorption due to increased surface uniformity, stability, and functional group availability.

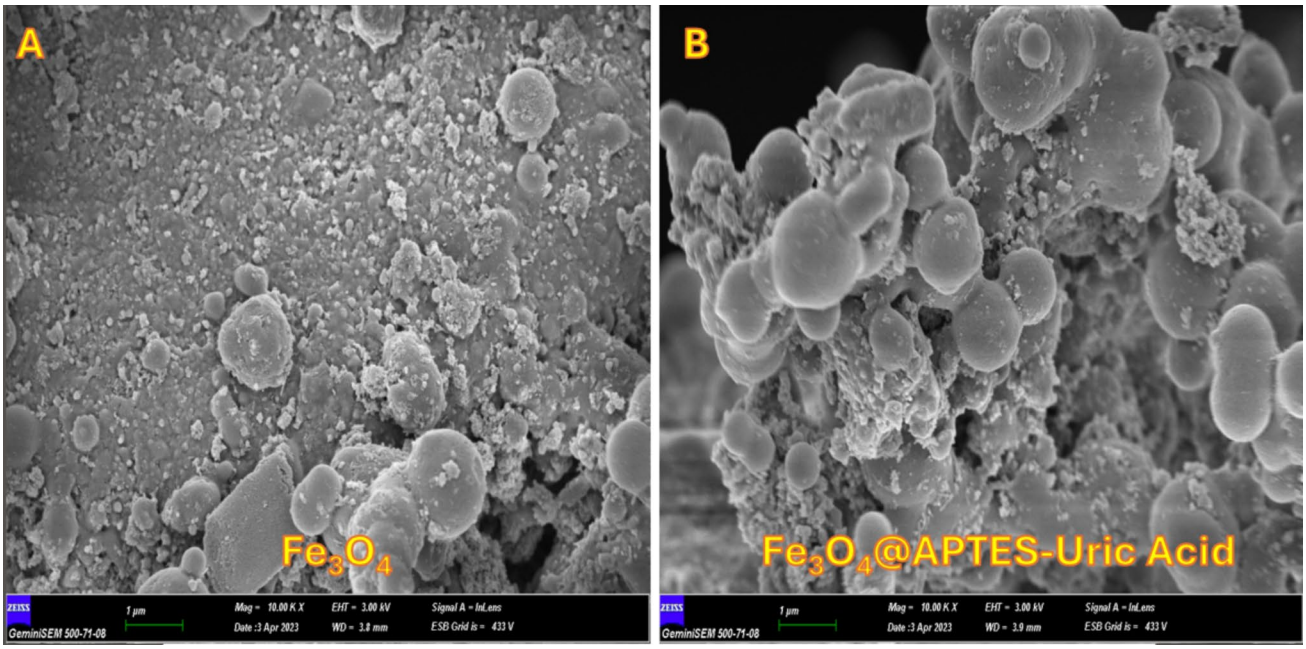


Fig. 1 SEM images of **A** Fe_3O_4 nanoparticles and **B** Fe_3O_4@APTES-uric acid composite. Bare Fe_3O_4 shows irregular aggregates with heterogeneous particle sizes and partial agglomeration. After APTES coating and uric acid functionalization, the material exhibits more uniform spherical structures with smoother surfaces, indicating successful modification. The morphological changes confirm the formation of a stable and functionalized adsorbent surface suitable for enhanced pesticide adsorption

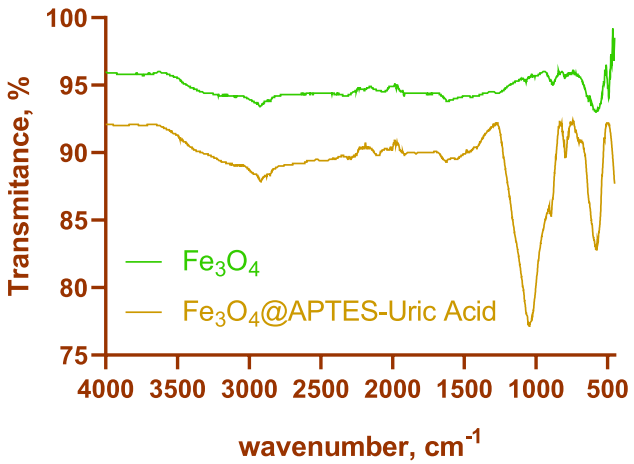


Fig. 2 FTIR spectra of bare Fe_3O_4 nanoparticles and the functionalized Fe_3O_4@APTES-uric acid nanocomposite

FTIR Analysis

The FTIR spectra of Fe_3O_4 and Fe_3O_4@APTES-uric acid confirm the successful surface functionalization of Fe_3O_4 nanoparticles. As can be seen in Fig. 2, the band observed around 574–579 cm^{-1} corresponds to the Fe–O stretching vibration, indicating the preserved magnetic core. Fe_3O_4 exhibits characteristic Si–O–Si stretching bands in the range of 1150–1247 cm^{-1}, confirming the presence of APTES on the surface (Nawaz et al. 2021). In contrast, Fe_3O_4@APTES-uric acid displays additional peaks at ~1650–1620 cm^{-1}, which can be attributed to C = O stretching and N–H bending vibrations derived from uric acid (Zeleňáková et al. 2024). Furthermore, the broad band around 3200–3500 cm^{-1}, assigned to O–H and N–H groups, becomes more intense in Fe_3O_4@APTES-uric acid, further verifying the incorporation of uric acid. These findings clearly demonstrate the effective conjugation of APTES and uric acid to the Fe_3O_4 nanoparticles (Yang et al. 2025).

EDS Analysis

The elemental composition of the Fe_3O_4@APTES-uric acid nanocomposite was investigated using the energy-dispersive X-ray spectroscopy (EDS), and the corresponding spectrum is shown in Fig. 3. As expected, intense signals at ~ 6.4 keV confirm the presence of Fe (Fe Kα), originating from the magnetite (Fe_3O_4) core. Strong oxygen (O Kα, ~ 0.52 keV) peaks were also detected, which are attributable to both the oxide lattice of Fe_3O_4 and the silica layer derived from APTES hydrolysis. In addition, the Si Kα signal at ~ 1.74 keV provides clear evidence of (Dawn et al. 2022; Roostaie et al. 2025) silica coating, indicating that the APTES-derived shell was successfully deposited on the magnetic nanoparticles. Several peaks assigned to Au are observed around 2 keV (M lines) and at higher energies near

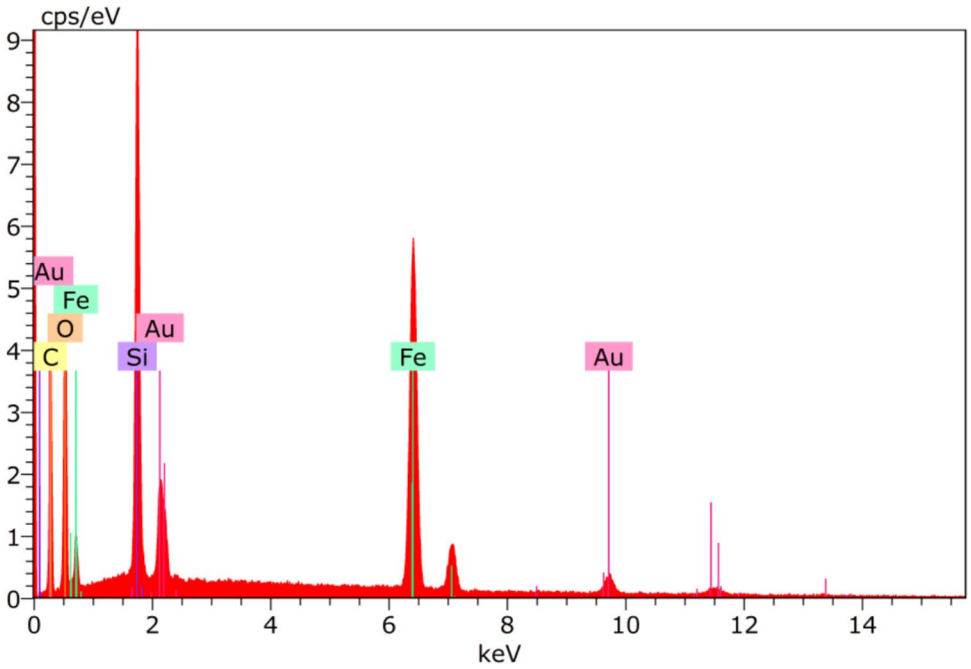


Fig. 3 X-ray spectrum of Fe_3O_4@APTES-uric acid magnetic sorbent

9.7–11 keV (L lines). These originate from the sputtered gold layer, which was applied to enhance surface conductivity during SEM/EDS measurements. Such signals are preparation artifacts and do not reflect the intrinsic composition of the sample. Carbon peaks (~0.28 keV) were also detected, which may arise from the carbon support tape and/or surface functionalization (Nawaz et al. 2020).

Although the material was further functionalized with APTES and uric acid, no distinct nitrogen peak was visible. This limitation is common in EDS, as the technique has inherently poor sensitivity for light elements (particularly N and H) and for thin organic surface layers. Therefore, the absence of detectable nitrogen does not exclude the successful immobilization of uric acid (Herbei et al. 2025; Lu et al. 2025). Complementary surface-sensitive techniques such as X-ray photoelectron spectroscopy (XPS) and Fourier transform infrared (FTIR) spectroscopy are recommended for confirming the presence and chemical state of organic ligands. In summary, the EDS spectrum verifies the coexistence of Fe, O, and Si, consistent with a Fe_3O_4@SiO_2 core–shell structure, while the Au signals originate from sputter coating. The results strongly support successful silica deposition, and the lack of N signal highlights the limitations of EDS for detecting surface-bound organics (Akhlaghi et al. 2020).

EDX Mapping Analysis

Figure 4 presents the EDX elemental mapping images of the synthesized Fe_3O_4 nanoparticles modified with APTES and uric acid. Panel A (red) corresponds to the distribution of Fe, confirming the homogeneous presence of the magnetic core. Panel B (green) shows the Si signal, indicating successful coating of the nanoparticles with APTES. Panel C (magenta) represents the N signal, verifying the introduction of nitrogen-rich functional groups from both APTES and uric acid. Panel D (blue) demonstrates the S distribution, originating from uric acid. The uniform and overlapping elemental signals confirm the effective surface modification and stable incorporation of functional groups.

Optimization of MSPE Experimental Studies

In the developed method, the separation and enrichment processes using the magnetic solid phase extraction (MSPE) technique were carried out under optimized conditions by systematically considering all critical parameters such as optimum pH, adsorption and desorption solvents and times, and other relevant factors. The aim of this study is to separate the target analyte components (fenarimol and pendimethalin) from other components in the medium by ensuring that they are adsorbed to the magnetic solid phase at the highest possible rate and to maintain that all target species adsorbed to the magnetic solid phase are transferred to the solvent after the separation process is completed. Unless otherwise stated, optimization experiments were carried out using model aqueous solutions spiked with 250 ng mL^{-1} of each analyte while keeping other parameters constant.

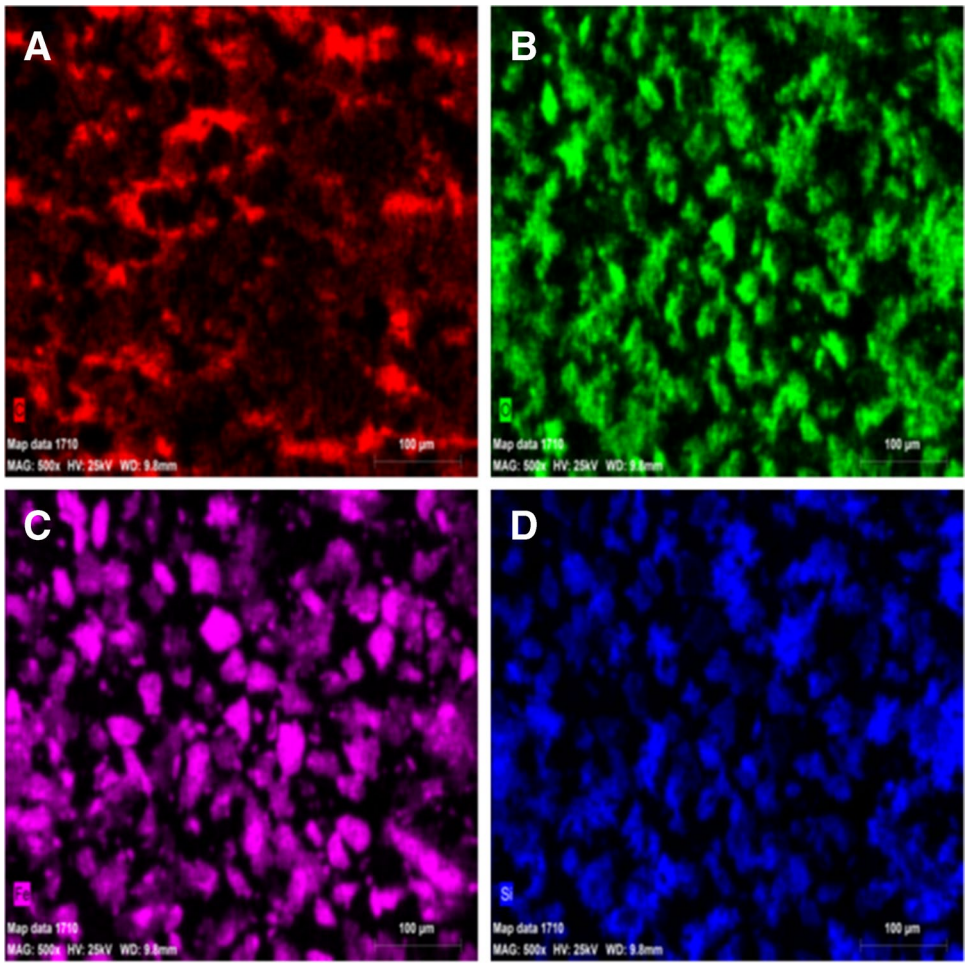


Fig. 4 EDX elemental mapping images of the synthesized Fe_3O_4 nanoparticles modified with APTES and uric acid. **A** Fe distribution (red) confirming the homogeneous presence of the magnetic core. **B** Si mapping (green) indicating successful APTES coating. **C** N signal (magenta) verifying the incorporation of nitrogen-containing functional groups from APTES and uric acid. **D** S distribution (blue) attributed to uric acid. The uniform and overlapping elemental distributions demonstrate effective surface modification of the magnetic nanoparticles

The Impact of pH on the Developed Method

A novel magnetic sorbent was synthesized for the trace extraction and analysis of the pesticides fenarimol and pendimethalin. All critical parameters of the magnetic solid-phase extraction (MSPE) protocol were subsequently optimized in a sequential manner. The initial optimization step focused on evaluating the influence of sample pH.

The key experimental variables governing the MSPE efficiency were optimized stepwise. First, the effect of pH on the extraction yield was studied using model aqueous solutions spiked with 250 ng mL^{-1} of each pesticide, across a pH range of 2 to 10. Consequently, it was prioritized as the first variable to be investigated by employing Britton-Robinson (BR) buffer solutions adjusted from pH 2 to 10. The pH of the medium plays a crucial role in the optimization process, as it governs both the ionization state of the analytes and their interactions with the magnetic phase. The pK_a values of the target pesticides are directly linked to their speciation: fenarimol, with a reported pK_a of approximately 2.6, exists predominantly in its ionized form (~96%) at pH 4.0, while pendimethalin, with a negligible pK_a value (2.80), remains essentially neutral across the entire studied pH range. Ionization strongly influences solubility and adsorption characteristics, since charged and neutral species interact differently with the magnetic surface. As shown in Fig. 5, both pesticides exhibited their maximum peak areas at pH 4.0. For fenarimol, this optimum can be attributed to the enhanced interaction of its anionic form with positively charged or hydrogen-bonding sites on the magnetic sorbent, thereby improving retention. In contrast, pendimethalin, which does not undergo significant ionization, likely achieves its maximum response at pH 4.0 due to favorable hydrophobic and surface interactions under these conditions. Overall, these results confirm that adjusting pH to values near 4.0 provides a balance that enhances extraction efficiency for both analytes, despite their distinct ionization behaviors.

In this study, after determining the ideal pH value after the optimization process, the effects of the pH volume to be added during the enrichment stage were investigated. The results (Figure S1) show that adding 2 mL of pH 4 buffer solution is appropriate.

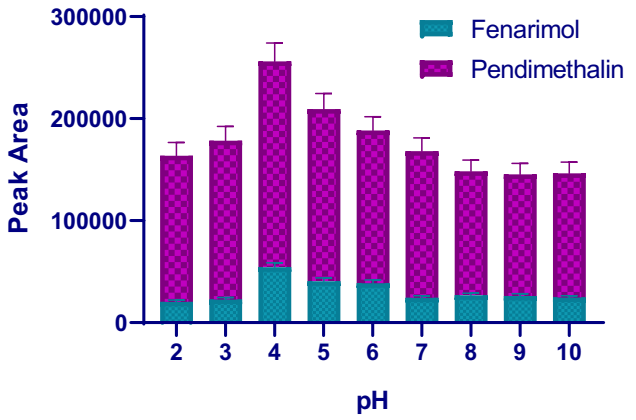


Fig. 5 Role of pH in the developed procedure (*n*, 3). Experimental conditions: sorbent (Fe_3O_4@APTES-uric acid) amount, 50 mg; sample volume, 50 mL (containing 250 ng mL^{-1} each of pendimethalin and fenarimol); adsorption time, 20 min. Eluent used, 1000 µL of ethanol. Desorption time, 30 s

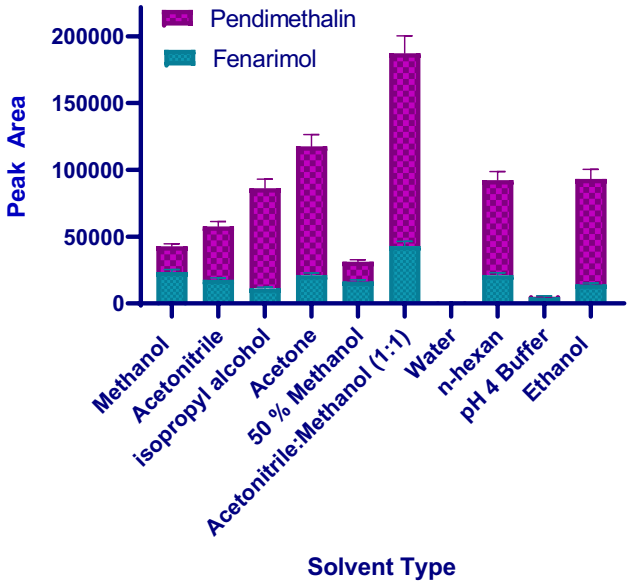


Fig. 6 Evaluation of desorption solvent for the proposed method (*n*, 3). Experimental conditions: sorbent (Fe_3O_4@APTES-uric acid) amount, 50 mg; pH of sample, 4.0; sample volume, 50 mL (containing 250 ng mL^{-1} each of pendimethalin and fenarimol); adsorption time, 20 min. Volume of eluents, 1000 µL. Desorption time, 30 s

The Impact of Eluent Selection for Back-Extraction

Desorption solution aims to effectively take the pendimethalin and fenarimol molecules adsorbed on the surface of the solid magnetic material into the liquid solution. For this purpose, the selection of a suitable solvent directly increases enrichment efficiency. Therefore, the correct selection of this eluent selection is one of the important optimization steps. Additionally, optimizing the extraction solution and volume increases the sensitivity of HPLC-DAD analysis results. This is especially important for the determination of low concentrations of pendimethalin and fenarimol analytes. In this regard, 10 separate tubes were used, and all parameters except the solvent were prepared to be the same. Methanol, acetonitrile, isopropanol, acetone, distilled water, n-hexane, pH: 4.0 BR buffer, acetonitrile:methanol (1:1), ethanol, and 50% methanol were tried as suitable solvents available in the laboratory. The most ideal peaks were reached with acetonitrile:methanol (1:1) solvent. The results obtained are shown in Fig. 6. It is seen that the most suitable solvent for the subsequent enrichment processes is the acetonitrile:methanol (1:1) solution.

The amount of desorption solvent will directly affect the enrichment factor because it determines the final volume of solution. Examining the volume of desorption solvent is one of the most important optimization steps. Using enough desorption solvent ensures effective separation of analytes from the surface of magnetic material. The volume of desorption solvent was also optimized, varying between 200 and 1500 µL. As can be seen in Fig. 7, the ideal peak signal was obtained with 500 µL of acetonitrile: methanol (1:1). It was observed that as the amount of solvent increased, the

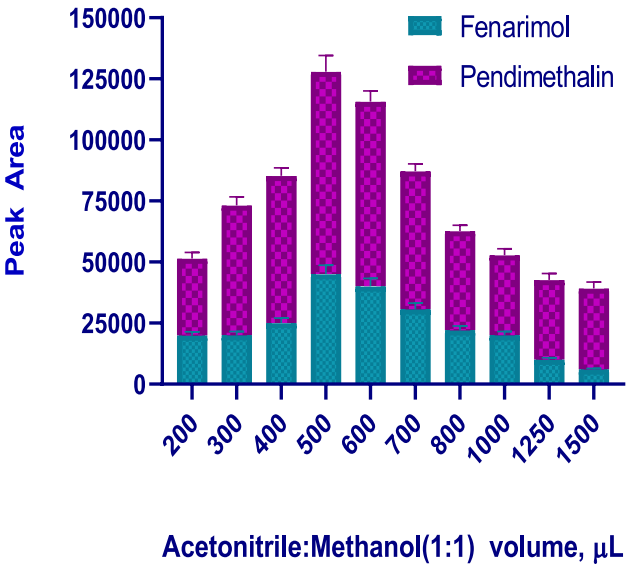


Fig. 7 Evaluation of the appropriate volume of ACN/MeOH for the proposed method (*n*, 3). Experimental conditions: sorbent (Fe_3O_4@APTES-uric acid) amount, 50 mg; pH of sample, 4.0; sample volume, 50 mL (containing 250 ng mL^{-1} each of pendimethalin and fenarimol); adsorption time, 20 min. Eluent for desorption, acetonitrile: methanol (1:1). Desorption time, 30 s

concentration of the analyte decreased, and the solution became diluted.

Effect of Time on Adsorption and Desorption Steps in MSPE

Adsorption time is the critical parameter for the complete retention of the target analytes on the magnetic materials. If sufficient contact time is not provided, the analytes cannot be fully adsorbed onto the magnetic material, which negatively affects the enrichment factor. Active sites on the surface of synthesized magnetic materials interact with target analytes. This interaction occurs through adsorption or chemical bonding. This step must be effectively optimized for both the enrichment factor and accurate and sensitive analysis. The adsorption process was carried out by using a shaker device at a constant rate. The effect of time was studied in the range of 5–120 min, and the optimal values for both pesticide molecules were reached with 40 min as shown in Fig. 8(A).

Following aqua phase from magnetic material, a desorption solvent is added to take target molecules from sorbent. A mechanical effect provided by a vortex increases the interaction between the solvent and sorbent, allowing for better desorption of the molecules. Therefore, the optimization of the vortex time for pendimethalin and fenarimol molecules needs to be done accurately and precisely. To determine the vortexing time, which is effective on the desorption amount of the analyte in the solvent, this parameter was scanned in the range of 5 and 120 s. As can be seen in Fig. 8(B), the most ideal peaks were obtained and determined as 60 s.

Reusability and Stability of the Magnetic Solid Phase

Optimization is of great importance to evaluate the stability of the synthesized magnetic material throughout the experimental period. Encouraging environmentally friendly practices, minimizing or eliminating the use of toxic solvents, and economically sustainable studies make the reuse of the original synthesized materials advantageous. Therefore, determining how many times magnetic material can be effectively used during the experimental period is one of the critical parameters of the study. In this study, experiments conducted using model solutions revealed no significant differences in the results even after five optimization processes. If the value measured during the first use was 100, the calculated values for each subsequent use were presented in Fig. 9.

Analytical Validation Parameters of the Proposed Method

After determining the most suitable experimental conditions for the MSPE, partly analytical validation of the proposed method was carried out by considering ICH guidelines and previously published articles (ICH 2005; Sharif et al. 2022; S. Ulusoy et al. 2024). To clearly demonstrate the enhancement provided by the MSPE step, performance was benchmarked against direct HPLC-DAD analysis without preconcentration (Before MSPE). The MSPE experiments were performed on fenarimol and pendimethalin solutions at different concentrations to determine the

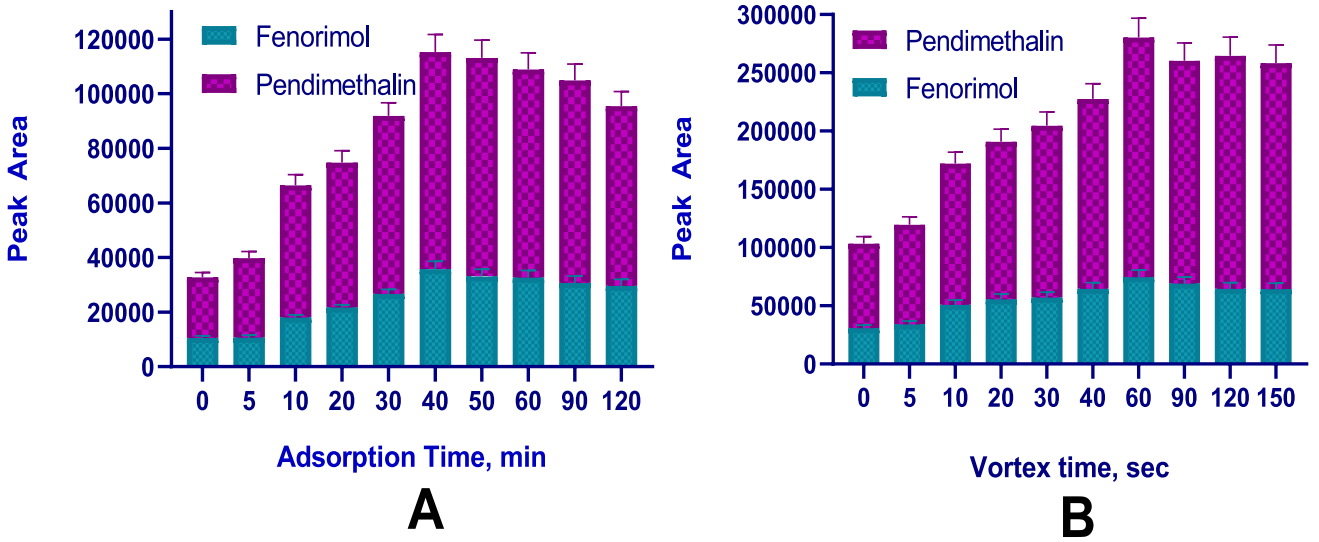


Fig. 8 Effect of time on adsorption (**A**) and desorption (**B**) steps in MSPE (*n*, 3). **A** Experimental conditions: sorbent (Fe_3O_4@APTES-uric acid) amount, 50 mg; pH of sample, 4.0; and sample volume, 50 mL (containing 250 ng mL^{-1} each of pendimethalin and fenarimol). Eluent for desorption, 500 µL of acetonitrile:methanol (1:1). Desorption time, 30 s. **B** Experimental conditions: sorbent (Fe_3O_4@APTES-uric acid) amount, 50 mg; pH of sample, 4.0; sample volume, 50 mL (containing 250 ng mL^{-1} each of pendimethalin and fenarimol); adsorption time, 40 min. Eluent for desorption, 500 µL of acetonitrile:methanol (1:1)

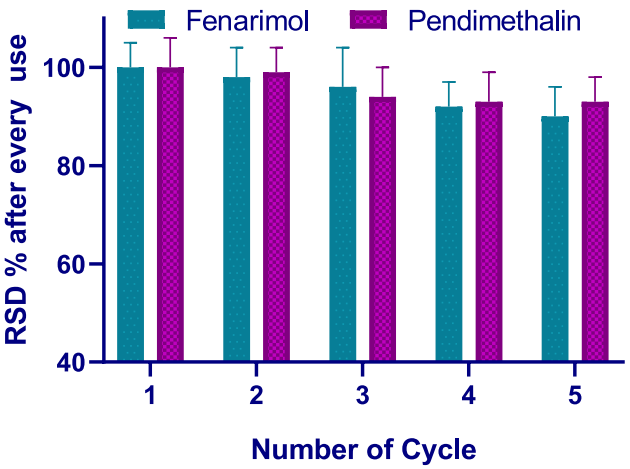


Fig. 9 Reusability durability of the magnetic material. Experimental conditions: sorbent (Fe_3O_4@APTES-uric acid) amount, 50 mg; pH of sample, 4.0; sample volume, 50 mL (containing 250 ng mL^{-1} each of pendimethalin and fenarimol); adsorption time, 40 min. Eluent for desorption, 500 µL of acetonitrile:methanol (1:1). Desorption time, 60 s

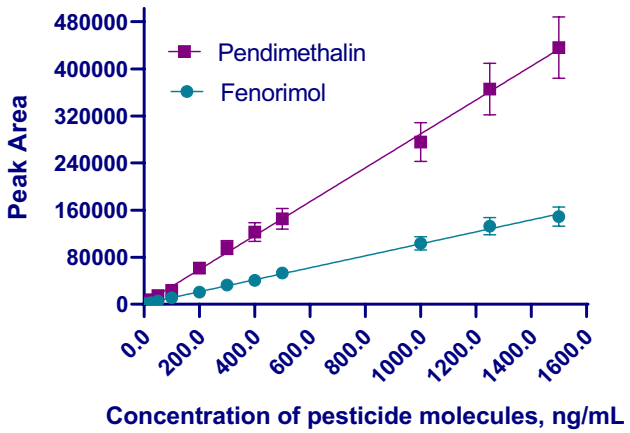


Fig. 10 Calibration curve after the method developed for fenarimol and pendimethalin

linear working range. According to the results, it was determined that the linearity in the concentration range of 10.00–1500.00 ng mL^{-1} for both pesticides was achieved after MSPE enrichment. The calibration curves obtained by applying the developed method are shown in Fig. 10 for fenarimol and pendimethalin, respectively. In addition, as seen after enrichment, the signals increased proportionally with concentration. Analytical parameters including limit of detection (LOD), limit of quantification (LOQ), enrichment factor, and recovery were calculated as stated in the ICH guidelines. For a direct comparison, the LOD and LOQ values from the Before MSPE calibration are also included alongside the After MSPE results in Table 3, quantitatively highlighting the sensitivity gain achieved.

Analysis of Trace-Level Fenarimol and Pendimethalin Using MSPE-HPLC-DAD in Fruit Juices

The developed method was applied to juice samples which are bought at local markets. The samples were subjected to analysis using both direct measurement and the standard addition method, in which the pesticide molecules were spiked at three different concentration levels to ensure accuracy and reliability of the results. The reproducibility of the results was evaluated based on analyses performed in triplicate and expressed as the relative standard deviation (RSD). In addition, the calculated recovery values provided detailed information regarding the accuracy of the developed method.

Five types of fruit juices were centrifuged at 3000 rpm for 5 min to remove particulates, and 10 mL of the clarified supernatant was collected. Aliquots were spiked with standard solutions of fenarimol and pendimethalin at three concentration levels (100.00, 250.00, 500.00 ng mL^{-1}) and analyzed in triplicate using the proposed method.

As shown in Table 4, recoveries ranged from 90.5% to 107.1%, while RSD values remained below 7.4% across all samples. These results demonstrate excellent accuracy, precision, and reproducibility, meeting accepted analytical criteria. The method's consistent recoveries and low variability further indicate minimal matrix effects, confirming its suitability for reliable quantification of the target analytes in complex juice matrices.

Evaluation of the Comprehensive and Practical Aspects of the Developed Analytical Method Using BAGI and MoGAPI Indices

The Blue Applicability Index (BAGI) is a novel scoring system (25–100 scale) developed to assess the practical viability of analytical methods (Fig. 11). It integrates key operational factors such as analytical throughput, automation potential, complexity, and reagent demands, thereby providing a standardized measure of methodological sustainability in routine laboratory practice (Manousi et al. 2023). Unlike conventional green chemistry metrics, which emphasize environmental impact (e.g., waste generation and toxicity), BAGI evaluates the feasibility of real-world implementation. As shown in Fig. 11a, the method achieved a BAGI score of 72.5 (Figure S2), suggesting overall suitability while highlighting the need for further practical optimization.

The Metric for Green Analytical Procedures Index (MoGAPI) was employed to assess the environmental impact of the developed MSPE-HPLC-DAD method for

Table 3 Analytical performance criteria of the proposed method ($n = 3$)

Parameter	Directly HPLC–DAD analysis of standard solutions (without MSPE)		With MSPE (via developed method)	
	Fenarimol	Pendimethalin	Fenarimol	Pendimethalin
Linear range	2.0–50.0 μg mL^{-1}	2.0–50.0 μg mL^{-1}	10.0–1500.0 ng mL^{-1}	10.0–1500.0 ng mL^{-1}
LOD[a]	0.78 μg mL^{-1}	0.84 μg mL^{-1}	3.22 ng mL^{-1}	3.32 ng mL^{-1}
LOQ[b]	1.89 μg mL^{-1}	1.92 μg mL^{-1}	9.52 ng mL^{-1}	9.90 ng mL^{-1}
RSD (%)	4.1	4.4	5.2	5.8
Calibration sensitivity	7.1	12.5	667.6	1075.1
Correlation coefficient (R^2)	0.9971	0.9989	0.9954	0.9973
Enrichment factor[c]	-	-	100	100
Enhancement factor[d]	-	-	94	86

[a.,b]The limit of detection (LOD) was determined as three times the standard deviation of blank signals obtained from at least three replicate measurements, while the limit of quantification (LOQ) was calculated as ten times the standard deviation

[c]Enrichment factor was derived from the ratio of the initial aqueous phase volume (50 mL) to the final volume after enrichment (0.5 mL)

[d]The enhancement factor was obtained by comparing the slopes of the calibration curves before and after the enrichment process

Table 4 Application of the developed method on real samples

Samples	Added ng mL^{-1}	Found[a] (ng mL^{-1})		RSD[b]%		Recovery[c]%	
		Fenarimol	Pendimethalin	Fenarimol	Pendimethalin	Fenarimol	Pendimethalin
Fruit Juice 1	0	< LOD	< LOD	-	-	-	-
	100.0	96.5 ± 5.2	95.1 ± 6.1	5.4	6.4	96.5	95.1
	250.0	246.5 ± 12.5	240.8 ± 12.4	5.1	5.1	98.6	96.3
	500.0	496.5 ± 28.2	485.5 ± 26.1	5.7	5.4	99.3	97.1
Fruit Juice 2	0	< LOD	< LOD	-	-	-	-
	100.0	90.5 ± 4.6	96.8 ± 4.4	5.1	4.5	90.5	96.8
	250.0	261.2 ± 10.4	256.8 ± 14.8	4.0	5.8	104.5	102.7
	500.0	475.4 ± 35.4	508.7 ± 20.9	7.4	4.1	95.1	101.7
Fruit Juice 3	0	< LOD	< LOD	-	-	-	-
	100.0	104.6 ± 4.4	106.4 ± 7.1	4.5	6.7	104.6	106.4
	250.0	267.8 ± 11.9	239.7 ± 13.2	4.4	5.5	107.1	95.9
	500.0	482.9 ± 29.7	475.9 ± 26.7	6.2	5.6	96.6	95.2
Fruit Juice 4	0	< LOD	< LOD	-	-	-	-
	100.0	98.7 ± 3.5	102.1 ± 5.2	3.5	5.1	98.7	102.1
	250.0	237.5 ± 16.4	238.4 ± 16.7	6.9	7.0	95.0	95.4
	500.0	465.8 ± 20.4	516.7 ± 21.8	4.4	4.2	93.2	103.3
Fruit Juice 5	0	< LOD	< LOD	-	-	-	-
	100.0	106.4 ± 3.8	98.6 ± 4.2	3.6	4.3	106.4	98.6
	250.0	254.8 ± 10.6	259.7 ± 13.6	4.2	5.2	101.9	103.9
	500.0	503.4 ± 24.7	479.6 ± 24.7	4.9	5.2	100.7	95.9

[a]Mean ± standard deviation (n, 3)

[b]RSD (%) was calculated from triplicate analyses of spiked samples under optimized conditions using the formula: RSD (%) = (SD/Mean) × 100

[c]Recovery (%) was determined by comparing the measured concentration after the MSPE procedure with the fortified concentration using the equation: Recovery (%) = (C_found/C_spiked) × 100

the determination of fenarimol and pendimethalin. MoGAPI considers ten key factors, including energy consumption, waste generation, reagent toxicity, and operator safety, to provide a comprehensive greenness score (Mansour et al. 2024). Our method achieved a score of 80 (Figure S3) (Fig. 11b), reflecting strong compliance with sustainable analytical practices. Favorable ratings in the "Reagents" and "Waste" categories were attributed to the low solvent

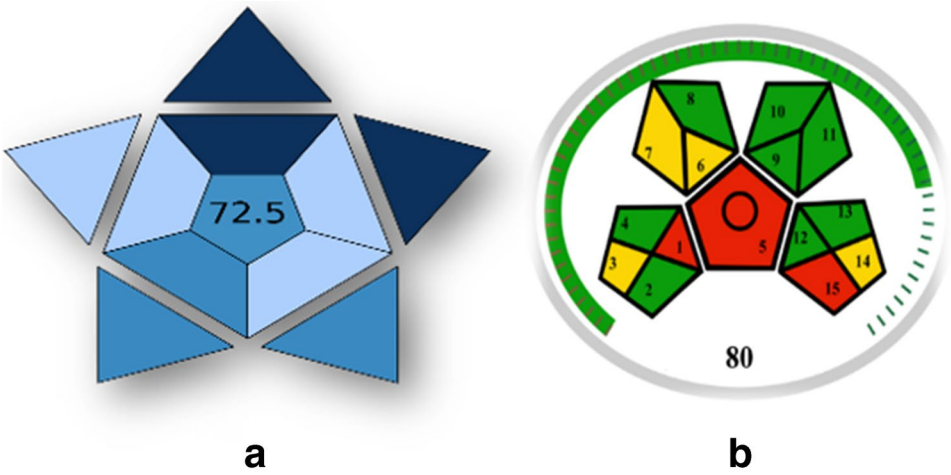


Fig. 11 Results obtained for green evaluation of the proposed method. **a** BAGI's pictogram system. **b** MoGAPI pictogram system

requirement (< 10 mL per sample) and room-temperature extraction. Additional reductions in environmental impact were achieved through the elimination of derivatization, energy-efficient HPLC conditions, and the use of a reusable magnetic sorbent in the MSPE step, which minimized cartridge waste.

Conclusion

In this study, a new magnetic adsorbent was synthesized for the high-sensitivity determination of trace levels of fenarimol and pendimethalin pesticide residues using a conventional HPLC-DAD system, and an MSPE-based separation and enrichment strategy were developed using this adsorbent. In addition, pesticide residues that typically require derivatization and GC-based analysis could be successfully determined using the proposed method with a conventional HPLC system. This eliminates additional sample treatment steps while simplifying the analytical workflow. Therefore, the developed approach offers a more practical and accessible alternative for routine pesticide monitoring. The developed MSPE method offers significant advantages compared to conventional solid-phase extraction (SPE), liquid-liquid extraction (LLE), and precipitation methods commonly used for pre-enrichment, thanks to the ease of magnetic separation, high surface area, and tunable functional groups of magnetic nanoparticles. The synthesized magnetic adsorbents exhibit high affinity for target pesticides, enabling effective interaction and isolation even in complex matrices. This method, which stands out in terms of low solvent consumption, short processing time, and the practical advantages of magnetic separation, constitutes a sustainable and effective alternative to conventional pre-enrichment techniques. Its alignment with green chemistry principles was quantitatively validated by a BAGI (Blue Applicability Green Index) score of 72.5 and a MoGAPI (Metric for Green Analytical Procedures Index) value of 80, confirming its sustainability, eco-friendliness, and minimal waste generation. The MoGAPI score (72.5) demonstrates that the proposed method provides a balanced profile in terms of analytical performance, practical applicability, and environmental impact, supporting its suitability for routine use. Furthermore, the environmental and operational greenness of the proposed procedure was assessed using the BAGI index. The obtained BAGI score supports that the method possesses a favorable green profile, considering reagent toxicity, waste generation, energy consumption, and procedural simplicity. This result further confirms that the developed analytical approach combines reliable analytical performance with environmentally responsible design. These results highlight the advantages of low solvent consumption (< 10 mL per sample), reusable magnetic sorbent, energy-efficient conditions, and the use of non-toxic reagents, making the developed method a promising green alternative for pesticide residue analysis in food matrices.

To our knowledge, this is one of the important studies using an MSPE-HPLC-DAD approach for the analysis of fenarimol and pendimethalin with its main strengths such as original adsorbent design, high selectivity, and user-friendly application. The use of a low-cost, fast, and easy magnetic separation system increases its potential for routine analysis and field applications. This study makes significant contributions to both academic and industrial research by providing a highly efficient, accessible, and environmentally friendly alternative for pesticide residue analysis.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s12161-026-03067-y.

Acknowledgements This study was partly supported by the project numbered 1919B012111760 within the scope of TÜBİTAK 2209-A University Student Research. Moreover, this study was produced from experimental data obtained from Dolunay Kızıltoprak's thesis. The authors would like to express their sincere gratitude to Dr. Gökhan Sarp for his valuable contributions to the characterization studies of the

synthesized magnetic material. His expertise and insightful discussions significantly enriched the experimental part of this work and are deeply appreciated. Language editing support was provided by ChatGPT (OpenAI), which was used to improve the linguistic accuracy and fluency of selected English sections of the manuscript. No content was generated or interpreted by the tool.

Author Contribution Songül ULUSOY; She prepared and wrote the manuscript Dolunay KIZILTOPRAK; She did some of the experiments. Esra DURGUN; She did some of the experiments Ümmügülsüm POLAT; she calculated analytical merits Marcello LOCATELLİ; He designed and composed of the manuscript HALİL İBRAHİM ULUSOY; He added research funding and evaluated data.

Funding Open access funding provided by the Scientific and Technological Research Council of Türkiye (TÜBİTAK).

Data Availability The data presented in this study is available in article or supplementary material here.

Declarations

Competing interest The authors declare no competing interests.

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